

PE-EC603D — Information Theory and Coding

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Module 1 Exam Notes: Basics & Source Coding

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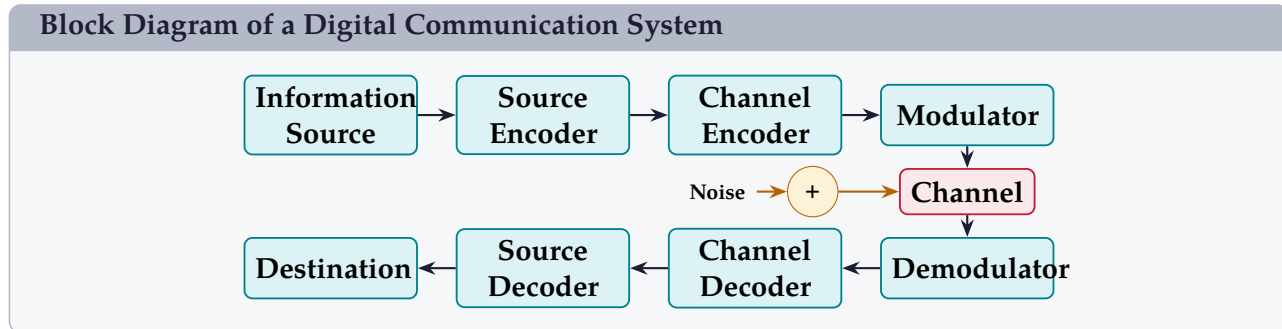
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1. Basics of Information Theory

Information theory is a mathematical framework developed by Claude Shannon in 1948 to quantify, store, and communicate digital information. It defines limits on data compression and transmission rates.

1.1 Model of a Digital Communication System

Before quantifying information, we analyze the basic blocks involved in its transmission:



1.2 Measure of Information Content

The "information" in a message is not related to its semantic meaning, but rather to its **uncertainty** or **surprise value**. An event that is highly likely to happen carries very little information when it occurs, whereas an extremely rare (unlikely) event carries high information.

Information Content: The quantitative measure of the uncertainty associated with the occurrence of an event. For a discrete source emitting a symbol x_i with probability $P(x_i)$, the information content $I(x_i)$ is inversely proportional to $P(x_i)$.

Information Measurement Formula

The information content $I(x_i)$ of symbol x_i is defined as:

$$I(x_i) = \log_r \left(\frac{1}{P(x_i)} \right) = -\log_r P(x_i) \quad (1)$$

The unit of measurement depends on the base r of the logarithm:

Base $r = 2$: Units are **Bits** (Binary digits). $I(x_i) = -\log_2 P(x_i)$.

Base $r = e$: Units are **Nats** (Natural units). $I(x_i) = -\ln P(x_i)$.

Base $r = 10$: Units are **Hartleys** (or Decits). $I(x_i) = -\log_{10} P(x_i)$.

Conversion Relations:

$$1 \text{ Nat} = \log_2 e \approx 1.443 \text{ Bits} \quad (2)$$

$$1 \text{ Hartley} = \log_2 10 \approx 3.322 \text{ Bits} \quad (3)$$

1.3 Entropy for Discrete Ensembles

A discrete information source emits symbols from an alphabet $X = \{x_1, x_2, \dots, x_M\}$ with corresponding probabilities $P = \{P(x_1), P(x_2), \dots, P(x_M)\}$, where $\sum_{i=1}^M P(x_i) = 1$.

Entropy $H(X)$: The average information content per symbol emitted by the source. It represents the mathematical expectation of the information of all symbols, measuring the overall uncertainty of the source.

Entropy Formula

$$H(X) = E[I(x_i)] = \sum_{i=1}^M P(x_i) I(x_i) = - \sum_{i=1}^M P(x_i) \log_2 P(x_i) \quad (\text{bits/symbol}) \quad (4)$$

1.4 Properties of Entropy

Entropy exhibits several fundamental properties that reflect its nature as a measure of uncertainty:

1. **Non-negativity:** $H(X) \geq 0$. Since $0 \leq P(x_i) \leq 1$, the term $-P(x_i) \log_2 P(x_i)$ is always non-negative.
2. **Certainty (Minimum Entropy):** $H(X) = 0$ if and only if $P(x_i) = 1$ for some symbol x_i , and $P(x_j) = 0$ for all $j \neq i$. In this case, there is no uncertainty in the source output.
3. **Maximality (Maximum Entropy):** $H(X) \leq \log_2 M$, where M is the number of symbols in the alphabet. The maximum uncertainty occurs when all symbols are equiprobable, i.e., $P(x_i) = 1/M$ for all i .

1.4.1 Derivation of Maximum Entropy Condition (Proof of Maximality)

To find the probability distribution that maximizes $H(X)$ subject to the constraint $\sum_{i=1}^M P(x_i) = 1$, we use the method of Lagrange multipliers. Let the objective function be:

$$f(P_1, P_2, \dots, P_M) = - \sum_{i=1}^M P_i \ln P_i$$

Subject to the constraint:

$$g(P_1, P_2, \dots, P_M) = \sum_{i=1}^M P_i - 1 = 0$$

Define the Lagrangian function J :

$$J = - \sum_{i=1}^M P_i \ln P_i - \lambda \left(\sum_{i=1}^M P_i - 1 \right)$$

Differentiating J with respect to P_i and setting it to 0 for maximum value:

$$\begin{aligned}\frac{\partial J}{\partial P_i} &= -\ln P_i - 1 - \lambda = 0 \\ \ln P_i &= -(1 + \lambda) \\ P_i &= e^{-(1+\lambda)}\end{aligned}$$

Since the right-hand side is constant and does not depend on the index i , all P_i must be equal:

$$P_1 = P_2 = \dots = P_M = C \quad (\text{a constant})$$

Using the constraint $\sum_{i=1}^M P_i = 1$:

$$M \times C = 1 \implies C = \frac{1}{M} \implies P_i = \frac{1}{M} \quad \text{for all } i.$$

Substituting $P_i = 1/M$ back into the entropy formula yields the maximum entropy:

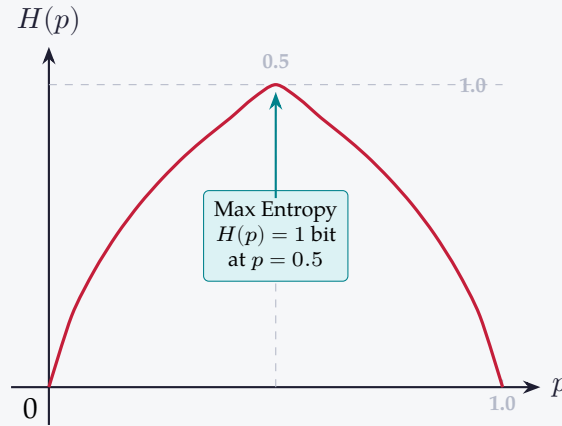
$$H(X)_{\max} = -\sum_{i=1}^M \frac{1}{M} \log_2 \left(\frac{1}{M} \right) = -M \left(\frac{1}{M} \right) (-\log_2 M) = \log_2 M \quad (\text{Q.E.D.})$$

1.5 Binary Entropy Function

For a binary source with alphabet $X = \{0, 1\}$ and probabilities $P(0) = p$ and $P(1) = 1 - p$, the entropy is denoted by $H(p)$ or $H_b(p)$:

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p) \quad (5)$$

Binary Entropy Function $H(p)$ vs p



2. Joint, Conditional, and Mutual Information

To analyze communication systems with an input source X and an output destination Y , we define joint and conditional entropy metrics over the joint ensemble (X, Y) .

2.1 Definitions of Entropies

Let $P(x_i)$ be input probabilities, $P(y_j)$ be output probabilities, and $P(x_i, y_j)$ be the joint probability of transmitting x_i and receiving y_j .

Joint and Conditional Entropies

1. **Joint Entropy** $H(X, Y)$: The total average uncertainty of the combined system.

$$H(X, Y) = - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 P(x_i, y_j) \quad (6)$$

2. **Conditional Entropy** $H(X|Y)$: The average uncertainty about the input X after the output Y is observed (also called **Equivocation**).

$$H(X|Y) = - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 P(x_i|y_j) \quad (7)$$

3. **Conditional Entropy** $H(Y|X)$: The average uncertainty about the output Y when the input X is known (represents the noise introduced by the channel).

$$H(Y|X) = - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 P(y_j|x_i) \quad (8)$$

2.1.1 Derivation of Joint Entropy Relations

We derive the relation $H(X, Y) = H(X) + H(Y|X)$ using joint and conditional probability relationships $P(x_i, y_j) = P(x_i)P(y_j|x_i)$:

$$\begin{aligned} H(X, Y) &= - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 P(x_i, y_j) \\ &= - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 [P(x_i)P(y_j|x_i)] \\ &= - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) [\log_2 P(x_i) + \log_2 P(y_j|x_i)] \\ &= - \sum_{i=1}^M \left(\sum_{j=1}^N P(x_i, y_j) \right) \log_2 P(x_i) - \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 P(y_j|x_i) \end{aligned}$$

Using the probability marginalization rule $\sum_{j=1}^N P(x_i, y_j) = P(x_i)$:

$$H(X, Y) = - \sum_{i=1}^M P(x_i) \log_2 P(x_i) + H(Y|X) = H(X) + H(Y|X) \quad (\text{Q.E.D.})$$

Similarly, using $P(x_i, y_j) = P(y_j)P(x_i|y_j)$, we can prove:

$$H(X, Y) = H(Y) + H(X|Y) \quad (9)$$

2.2 Mutual Information

Mutual Information $I(X; Y)$: The amount of information about the source X that is obtained by observing the channel output Y . It measures the reduction in uncertainty of X due to the knowledge of Y .

Mutual Information Formulas

$$I(X; Y) = H(X) - H(X|Y) \quad (\text{bits/symbol}) \quad (10)$$

By symmetry and joint entropy relations, it can also be expressed as:

$$I(X; Y) = H(Y) - H(Y|X) \quad (11)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (12)$$

In terms of probabilities:

$$I(X; Y) = \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) \log_2 \left[\frac{P(x_i, y_j)}{P(x_i)P(y_j)} \right] \quad (13)$$

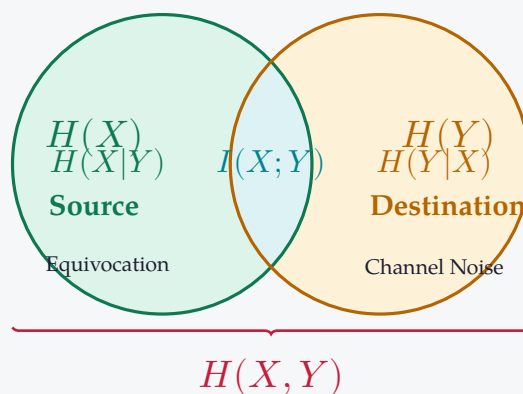
2.3 Properties of Mutual Information

- **Symmetry:** $I(X; Y) = I(Y; X)$. The channel output Y carries the same amount of information about the input X as the input X carries about the output Y .
- **Non-negativity:** $I(X; Y) \geq 0$. The observation of Y can never increase the uncertainty of X on average.
- **Independence Condition:** $I(X; Y) = 0$ if and only if X and Y are statistically independent. In this case, receiving Y provides no information about X .
- **Relation to Channel Capacity:** For deterministic, noiseless channels, conditional uncertainty vanishes, and $I(X; Y) = H(X)$.

2.4 Entropy Relationship Diagram

The relationships between all these entropy measures are summarized in the Venn diagram below:

Venn Diagram of Entropy Relationships



3. Shannon's First Theorem & Source Coding

Source coding is the process of representing discrete source symbols with binary code words. The goal is to reduce data size by removing redundancy.

3.1 Shannon's Noiseless Coding Theorem

Shannon's First Theorem (Source Coding Theorem): For a discrete memoryless source X with entropy $H(X)$ (bits/symbol), the average codeword length L (bits/symbol) for any uniquely decodable binary code is bounded by:

$$L \geq H(X) \quad (14)$$

By coding sequences of N symbols at a time (block coding), the average code length per source symbol L_N can be made arbitrarily close to $H(X)$ as $N \rightarrow \infty$:

$$\lim_{N \rightarrow \infty} L_N = H(X) \quad (15)$$

3.2 Performance Metrics of Source Codes

For a source emitting symbols s_i with probability $P(s_i)$ and assigned codeword length l_i (in bits):

Average Length, Efficiency, and Redundancy

1. Average Codeword Length (L):

$$L = \sum_{i=1}^M P(s_i) l_i \quad (\text{bits/symbol}) \quad (16)$$

2. Coding Efficiency (η):

$$\eta = \frac{H(X)}{L} \times 100\% \quad (17)$$

3. Code Redundancy (R):

$$R = 1 - \eta = 1 - \frac{H(X)}{L} \quad (18)$$

3.3 Prefix Codes and Kraft-McMillan Inequality

A code is **uniquely decodable** if any concatenated sequence of codewords can be decoded in only one way.

Prefix Code (Instantaneous Code): A uniquely decodable code with the property that no codeword is a prefix of any other codeword. This allows a sequence to be decoded symbol-by-symbol instantaneously as it is received, without waiting for the end of the sequence.

Kraft-McMillan Inequality

A prefix code (or any uniquely decodable code) with alphabet size D (usually binary, $D = 2$) and codeword lengths $l_1, l_2, \dots, l_M$ exists if and only if:

$$\sum_{i=1}^M D^{-l_i} \leq 1 \quad (19)$$

For binary codes, this is:

$$\sum_{i=1}^M 2^{-l_i} \leq 1 \quad (20)$$

If the inequality is satisfied with equality ($\sum 2^{-l_i} = 1$), the code is called a **complete code** (no further codewords can be added without violating the prefix property).

3.4 Shannon-Fano Coding Algorithm

Shannon-Fano coding is a top-down algorithm to construct prefix codes. The procedure is:

- Step 1:** Sort the source symbols in descending order of their probabilities.
- Step 2:** Divide the sorted list of symbols into two subsets such that the sum of probabilities of the two subsets is as close to equal as possible.
- Step 3:** Assign binary code '0' to all symbols in the first (upper) subset and '1' to all symbols in the second (lower) subset.
- Step 4:** Recursively apply Steps 2 and 3 to each subset until each sub-sublist contains exactly one symbol.
- Step 5:** The codeword for each symbol is the sequence of 0s and 1s assigned to it from the root of divisions.

Worked Example: Shannon-Fano Coding

Problem: A discrete memoryless source emits 5 symbols $\{s_1, s_2, s_3, s_4, s_5\}$ with probabilities $P = \{0.35, 0.30, 0.20, 0.10, 0.05\}$.

- a. Construct a binary Shannon-Fano code.
- b. Calculate the entropy $H(X)$ of the source.
- c. Calculate the average codeword length L and the efficiency η .

Solution:

(a) Code Construction Table:

Symbol	Prob.	Stage 1	Stage 2	Stage 3	Codeword	Length (l_i)
s_1	0.35	0	0		00	2
s_2	0.30	0	1		01	2
s_3	0.20	1	0		10	2
s_4	0.10	1	1	0	110	3
s_5	0.05	1	1	1	111	3

Division Rationale:

- **Stage 1:** Split between $\{s_1, s_2\}$ (sum = 0.65) and $\{s_3, s_4, s_5\}$ (sum = 0.35). Difference is 0.30. (Splitting between s_1 and others yields difference $0.35 - 0.65 = 0.30$, so both partitions are valid. Let's use the first).
- **Stage 2:** Split upper subset $\{s_1\}$ (0.35) and $\{s_2\}$ (0.30). Split lower subset $\{s_3\}$ (0.20) and $\{s_4, s_5\}$ (0.15).
- **Stage 3:** Split $\{s_4\}$ (0.10) and $\{s_5\}$ (0.05).

(b) Source Entropy $H(X)$:

$$\begin{aligned}
 H(X) &= - \sum_{i=1}^5 P(s_i) \log_2 P(s_i) \\
 &= -[0.35 \log_2 0.35 + 0.30 \log_2 0.30 + 0.20 \log_2 0.20 + 0.10 \log_2 0.10 + 0.05 \log_2 0.05] \\
 &= -[0.35(-1.5146) + 0.30(-1.7370) + 0.20(-2.3219) + 0.10(-3.3219) + 0.05(-4.3219)] \\
 &= -[-0.5301 - 0.5211 - 0.4644 - 0.3322 - 0.2161] = 2.0639 \text{ bits/symbol}
 \end{aligned}$$

(c) Average Codeword Length L and Efficiency η :

$$\begin{aligned}
 L &= \sum_{i=1}^5 P(s_i) l_i = (0.35 \times 2) + (0.30 \times 2) + (0.20 \times 2) + (0.10 \times 3) + (0.05 \times 3) \\
 &= 0.70 + 0.60 + 0.40 + 0.30 + 0.15 = 2.15 \text{ bits/symbol}
 \end{aligned}$$

$$\eta = \frac{H(X)}{L} \times 100\% = \frac{2.0639}{2.15} \times 100\% \approx 95.99\%$$

$$\text{Redundancy } R = 1 - 0.9599 = 0.0401 \text{ or } 4.01\%$$

Kraft Inequality Verification:

$$\sum_{i=1}^5 2^{-l_i} = 2^{-2} + 2^{-2} + 2^{-2} + 2^{-3} + 2^{-3} = 0.25 + 0.25 + 0.25 + 0.125 + 0.125 = 1.0 \leq 1$$

The sum equals 1.0. This indicates that the code is prefix-free and is a complete code.

Quick Revision — Module I

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Key Formulas at a Glance

Self-Information	$I(x_i) = \log_2 \left(\frac{1}{P(x_i)} \right) = -\log_2 P(x_i)$
Entropy (Average Info)	$H(X) = -\sum_{i=1}^M P(x_i) \log_2 P(x_i)$
Joint Entropy	$H(X, Y) = -\sum \sum P(x_i, y_j) \log_2 P(x_i, y_j)$
Conditional Entropies	$H(X Y) = H(X, Y) - H(Y) \quad \text{and} \quad H(Y X) = H(X, Y) - H(X)$
Mutual Information	$I(X; Y) = H(X) - H(X Y) = H(Y) - H(Y X)$
Average Code Length	$L = \sum P(s_i) l_i$
Coding Efficiency	$\eta = \frac{H(X)}{L} \times 100\%$
Kraft Inequality	$\sum 2^{-l_i} \leq 1 \quad (\text{For binary prefix codes})$

One-Line Definitions

- **Information Content:** A measure of surprise or uncertainty of an event, dependent only on its probability.
- **Entropy:** The average uncertainty or information content per symbol emitted by a source.
- **Equivocation $H(X|Y)$:** The residual uncertainty of the input X after output Y is observed at the receiver.
- **Mutual Information:** The amount of information shared between the transmitter and receiver.
- **Prefix Code:** A code in which no codeword is a prefix of another, enabling instantaneous decoding.
- **Kraft Inequality:** A mathematical condition determining the existence of prefix codes for given lengths.
- **Source Coding Theorem:** Shannon's limit stating that average codeword length cannot be less than source entropy.
- **Redundancy:** The percentage of code space wasted by transmitting non-informative bits ($R = 1 - \eta$).

Frequently Asked / Viva Questions

Q1. State the units of information and their conversion relations.

Bits (base 2), Nats (base e), and Hartleys (base 10). Conversions: 1 Nat = 1.443 Bits; 1 Hartley = 3.322 Bits.

Frequently Asked / Viva Questions (contd.)

Q2. Under what condition is the entropy of a source maximized?

When all M symbols of the source alphabet are equiprobable, i.e., $P(x_i) = 1/M$ for all i . The maximum entropy is $H(X)_{\max} = \log_2 M$.

Q3. What is the physical significance of conditional entropy $H(Y|X)$?

It represents the average uncertainty of the output Y when the input X is known. Physically, it measures the noise added by the channel.

Q4. Explain the term "Equivocation" in communication channels.

Equivocation is $H(X|Y)$, which is the average uncertainty about the transmitted symbol X after receiving symbol Y . It represents lost information.

Q5. Why is Mutual Information symmetric?

$I(X;Y) = I(Y;X) = H(X) + H(Y) - H(X,Y)$. It represents the shared information between X and Y , making the direction of analysis irrelevant.

Q6. State Shannon's Source Coding Theorem.

The average length L of any uniquely decodable binary code for a source X satisfies $L \geq H(X)$. The minimum average code length is limited by the source entropy.

Q7. What is an instantaneous code (prefix code)?

A code where no codeword is a prefix of any other codeword. It is decodable immediately upon receiving the last bit of the codeword, without looking ahead.

Q8. What is the significance of the Kraft-McMillan Inequality?

It provides a necessary and sufficient condition for the existence of an instantaneous (prefix) code for a given set of codeword lengths.

Q9. If a source has an entropy of 1.8 bits/symbol and is encoded using a code with average length 2 bits/symbol, find efficiency and redundancy.

Efficiency $\eta = \frac{1.8}{2} \times 100 = 90\%$. Redundancy $R = 100\% - 90\% = 10\%$.

Q10. Why does block coding increase source coding efficiency?

Block coding groups N symbols together, allowing the average length per symbol to approach the entropy $H(X)$ more closely as $N \rightarrow \infty$ by utilizing joint probabilities.

Q11. Can the efficiency of a source code be greater than 100%?

No. According to Shannon's Source Coding Theorem, $L \geq H(X)$, meaning efficiency $\eta = H(X)/L \leq 1.0$ (or 100%).

Q12. State the main difference between Shannon-Fano and Huffman coding approaches.

Shannon-Fano is a top-down approach that partitions probabilities into subgroups, while Huffman coding is a bottom-up approach that recursively merges the lowest probabilities.

Comparison of Source Coding Techniques

Parameter	Fixed-Length Coding	Variable-Length Coding
Codeword Length	Equal length ($l_i = \lceil \log_2 M \rceil$) for all symbols.	Different lengths based on symbol probabilities.
Probability Sensitivity	Ignores probability distribution.	Allocates shorter codes to high-probability symbols.
Efficiency	Low if symbols have unequal probabilities.	High (approaches Shannon's limit $H(X)$).
Synchronization	Easy; boundaries are known.	Harder; requires prefix-free property (Prefix codes).
Examples	ASCII, Standard PCM.	Shannon-Fano, Huffman coding.